PROJECT GROUND-TRUTH DOCUMENT

NER Cognitive Gaming & Memory Assistance Platform

A localized, offline-first cognitive support platform for elderly users in North-East India

> **Scope statement.** This platform is not a dementia cure or diagnostic tool. Its immediate purpose is cognitive engagement, daily memory assistance, culturally familiar reminiscence, and caregiver monitoring. The more advanced AI components come later, once enough real usage data exists.

Target region: North-Eastern Region (NER), India
Primary users: Elderly user (mobile app) and Caregiver (web dashboard)
Team size: 6 people
Document status: Ground truth — reflects only what has been finalized. No unconfirmed detail added.

Contents

1. Project Overview

The core idea is to build a localized, offline-first cognitive support platform for elderly users in North-East India, especially people with mild cognitive impairment or early-stage dementia support needs.

The platform is not a dementia cure or diagnostic tool. Its immediate purpose is to help with cognitive engagement, daily memory assistance, culturally familiar reminiscence, and caregiver monitoring. The more advanced AI components come later, once enough real usage data exists. That framing matches the evidence and caution established in the research phase of this project.

What makes this project distinctive

The stronger story is not “we made dementia games.” It is: an elderly-focused cognitive support platform designed specifically around the realities of North-East India. Its differentiation comes from combining:

- Cognitive games
- Adaptive personalization
- Culturally localized reminiscence
- NER language support
- Offline-first operation
- Caregiver involvement
- Longitudinal behavioral data

This intersection is the gap not addressed by existing platforms.

2. Who Uses the System

There are mainly two users.

Elderly user

Uses the mobile app for:

- Cognitive games
- Reminders
- Family memories
- Culturally familiar reminiscence content
- Eventually, voice interaction

Caregiver

Uses a separate dashboard to:

- Monitor game participation
- See performance trends
- Create reminders
- Upload family photos / content
- See alerts or meaningful long-term changes

> **Clinician:** A clinician can be added later, but the clinician portal is not a core MVP requirement.

3. Cognitive Games

The elderly user gets short games that exercise different cognitive abilities. The design spans memory, attention, language, executive function, visuospatial ability and arithmetic.

Game type	What it exercises
Matching pairs	Memory
Tap / find target	Attention + reaction speed
Arrange steps	Executive function
Picture naming	Language / recall
Simple arithmetic	Calculation
Path / maze tasks	Visuospatial ability

MVP game selection

For the first MVP, build only 4 good games, not 8–12 mediocre ones:

1. Memory matching
2. Attention / reaction game
3. Sequencing game
4. Picture naming

4. Every Game Collects Performance Data

Every session should record:

- user_id
- game_type
- difficulty
- accuracy
- reaction_time
- errors
- hints_used
- session_duration
- completed_or_quit
- timestamp

This is important because the app is not only providing cognitive activities — it is also gradually building a longitudinal behavioral dataset. For example:

```
Week 1    Accuracy = 84%    Reaction = 2.8s
Week 4    Accuracy = 81%    Reaction = 3.0s
Week 10   Accuracy = 72%    Reaction = 4.1s
```

That allows us to study changes over time.

5. Adaptive Difficulty

For version 1, we do not use reinforcement learning yet. We use a straightforward rule-based staircase system:

```
 3 strong rounds   →  increase difficulty
 2 poor rounds     →  decrease difficulty
otherwise          →  keep same difficulty
```

This makes the game personalized from day one without needing training data. Later, once we accumulate real user telemetry, we can replace or augment this with reinforcement learning / contextual bandits.

6. Future Reinforcement-Learning System

Later, the RL system learns what difficulty or activity is appropriate for each user. Conceptually:

```
STATE   →   recent accuracy, reaction time,
         current difficulty, recent errors,
                 session behaviour
                        ↓
 ACTION  →   choose next difficulty / game
                        ↓
                   USER PLAYS
                        ↓
REWARD  →   did the user perform well without
                the task being too easy?
```

The goal is not to make the user score 100% — because then the system would just give easy games. The goal is to keep the person appropriately challenged and engaged. The Track B gameplay data is what eventually supports this personalization system.

7. Reminiscence Feature

This is one of the strongest parts of the project. Instead of generic images, the app contains culturally and personally familiar content:

- **Cultural:** Bihu, Hornbill Festival, regional foods, traditional clothing, local landmarks, regional songs
- **Personal:** family photographs, childhood places, old wedding pictures

Culturally localized reminiscence is a key opportunity for the NER context. The caregiver can also upload personal material. For example:

> **Photo:** Family wedding
>
> **Year:** 1998
>
> **People:** Daughter, son, husband
>
> **Location:** Guwahati

The app could then show prompts such as “Do you remember who this is?” or “Where was this photograph taken?” This creates a natural overlap between cognitive gaming, personal memories, and reminiscence.

8. Memory Assistance

The app also assists the user with everyday life.

Reminders

- Medicine at 8 PM
- Lunch at 1 PM
- Doctor appointment tomorrow

Reality orientation

> Wednesday
>
> 26 August 2026
>
> Good evening.
>
> You are at home.
>
> Next reminder: Medicine at 8 PM

Family recall

The caregiver can upload a photo, name, relationship, and short description. The elderly user can then revisit familiar people.

9. Caregiver Dashboard

The caregiver gets a normal web dashboard. They might see:

> **Sessions this week:** 5
>
> **Average memory accuracy:** 78%
>
> **Reaction time:** 3.2s
>
> **Games completed:** 12
>
> **Missed reminders:** 1

Along with trend graphs (e.g. memory accuracy over time). They can also:

- Create reminders
- Upload family photographs
- Manage reminiscence content
- See activity history

Later, meaningful performance deviations could generate alerts. The dashboard supports progress trends, alerts, medication logs and caregiver support.

10. Cognitive Trend Monitoring

This is different from reinforcement learning.

- **RL asks:** “What game / difficulty should I give this user next?”
- **Trend monitoring asks:** “Is this person’s performance changing over time?”

Initially we compute simple statistics:

- 7-day accuracy average and 30-day accuracy average
- Average reaction time and reaction-time change
- Error-rate change and game completion rate

Then compare current performance against the personal baseline. For example:

> **Memory-game response time has increased 18% compared with the user's normal baseline.**

We do not say: **"Dementia probability: 83%."** Not without proper clinical validation.

11. Long-Term Decline-Detection ML

Eventually this could become a genuine research component, drawing on gameplay telemetry, longitudinal data, and periodic validated clinical assessments. You could then study whether reaction time ↑, accuracy ↓, errors ↑, and task abandonment ↑ correlate with clinically assessed cognitive changes.

That is where supervised ML models such as XGBoost, Random Forest, or time-series models could potentially come in. But that is not v1.

12. Voice Assistant

Eventually we want the app to be voice-first, because typing and reading may be difficult for some elderly users.

```
          User speaks
               ↓
     ASR   (Speech → Text)
               ↓
       Assistant / backend
               ↓
          Response text
               ↓
     TTS   (Text → Speech)
               ↓
        User hears answer
```

Example — user: *"When is my medicine?"* System: *"Your evening medicine is scheduled for 8 PM."*

NER language support is one of the hardest but most valuable parts of the platform.

13. Language Scope

We should not attempt all NER languages initially. For MVP: one regional language first — Assamese is a sensible candidate. Potential later expansion:

```
Assamese  →  Manipuri / Meitei  →  Bodo  →  other regional languages
```

Starting with only a few languages is deliberate, because localization at scale is a major difficulty.

14. Speech-Data Research Track

If we eventually want genuinely good NER speech support, we may need our own small regional speech corpus. For example, 50–100 elderly Assamese speakers performing tasks such as:

- Picture description
- Object naming
- Simple conversation
- Short memory tasks

We would collect audio + human-verified transcription + language + task metadata, then use it to evaluate or fine-tune an existing pretrained multilingual model — not train a speech model from scratch.

> **Ethics:** This part requires proper ethics / consent if collected from real vulnerable participants.

15. Offline-First Design

This is an important architectural requirement, because connectivity may be unreliable in rural and hill regions. A naive design (App → Internet → Backend) means “no internet = broken app.” Instead:

```
                INTERNET?
               /         \
            YES            NO
             ↓              ↓
         FastAPI        SQLite
             ↑              |
             └─── Sync ────┘
```

So even offline, the user can play games, see cached memories, receive locally scheduled reminders, and generate gameplay results. When internet returns:

```
Phone  →  unsynced results  →  FastAPI  →  PostgreSQL
```

16. Technology Decisions

Mobile application — React Native

Chosen for Android/iOS cross-platform support, a good ecosystem, and suitability for an MVP; it handles mobile UI, notifications, and storage. The UI needs large buttons, simple navigation, high contrast, minimal text, and eventually voice-driven interaction.

Local mobile database — SQLite

SQLite sits directly on the user’s phone and stores cached games, reminders, basic user profile, recent results, cached reminiscence information, and pending sync events. This is what enables offline-first functionality.

Backend — FastAPI + Python

FastAPI handles all server communication. Example endpoints:

- POST /game-sessions
- GET /users/{id}/analytics
- POST /reminders · GET /reminders
- POST /memories
- GET /caregiver/{id}/patients

FastAPI is preferred largely because the later analytics / ML stack will also be Python.

Main database — PostgreSQL

Stores structured server-side data. Likely tables:

- users, caregivers, caregiver_user_links, games, game_sessions, reminders, memory_items, performance_metrics, alerts, sync_events

Media storage — object storage

Family photos, music and videos shouldn't be stored directly inside PostgreSQL. Use object storage such as AWS S3, Cloudflare R2, or Supabase Storage.

PostgreSQL stores	Object storage stores
photo name, user, description, URL	the actual .jpg / .mp3 / .mp4

Caregiver frontend — React / Next.js

The caregiver web dashboard. Both the React Native elderly app and the React/Next caregiver dashboard communicate with the same FastAPI backend.

17. Overall Architecture

```
                     ELDERLY USER
                          ↓
                  React Native App
                          |
        ┌─────────────┼─────────────┐
        ↓             ↓             ↓
      Games       Reminders      Memories
                      |
                      ↓
             Rule-based adaptation
                      |
                      ↓
           SQLite  (local / offline data)
                      |
                   |  sync
                      ↓
                   FastAPI
                      |
          ┌─────┴─────────────┐
          ↓                   ↓
      PostgreSQL       Object Storage
          |                   |
      gameplay         photos / audio
        telemetry           / video
                   |
                   ↓
            Analytics Engine
                   |
          ┌───┴──────────┐
          ↓              ↓
        Trends       Future RL / ML
                         |
                         ↓
     Caregiver Dashboard  (React / Next.js)
```

18. Final Tech Stack

Layer	Technology
Elderly mobile app	React Native
Caregiver dashboard	React / Next.js
Backend	FastAPI + Python
Main database	PostgreSQL
Offline mobile storage	SQLite
Media storage	S3 / R2 / Supabase Storage
API	REST / HTTPS

Layer	Technology
Initial adaptation	Rule-based staircase algorithm
Analytics	Python / Pandas / scikit-learn initially
Later personalization	Contextual bandits / RL
Later decline ML	XGBoost / other validated models
Speech	Existing multilingual ASR / TTS model or API initially
Later NER speech	Fine-tuned multilingual model
Notifications	Mobile local + push notifications

19. The Three Data Tracks

You don't choose between them. They are three things happening at different stages.

Track	Role	What it is
Track A	PRODUCT	Build now. No dataset. Games, reminders, reminiscence, rule-based adaptation.
Track B	BEHAVIORAL DATA	As users play: accuracy, reaction time, errors, difficulty, session length. Later powers personalization / RL and longitudinal cognitive research.
Track C	LANGUAGE / SPEECH DATA	If/when we want proper NER-language speech: collect consented NER speech, then evaluate / fine-tune a pretrained speech model.

20. MVP Scope (v1)

Elderly app

- Login / profile
- Assamese / English interface
- 4 cognitive games
- Rule-based difficulty adaptation
- Reminders
- Reality orientation
- Reminiscence gallery
- Family photo memories
- Offline storage

Backend

- Authentication / users
- Game-session logging
- Reminders API
- Memory-content API
- Offline synchronization
- Telemetry storage

Caregiver dashboard

- Link caregiver ↔ elderly user
- Create reminders
- Upload memories
- See game history
- Basic performance graphs

Analytics

- Accuracy trend
- Reaction-time trend
- Sessions / week
- Completion rate
- Personal baseline

21. What Is NOT in the MVP

Do not build these initially — they are extensions:

- Dementia diagnosis
- Clinical-risk percentage
- Full RL system
- Custom speech model
- Speech biomarkers
- Full doctor dashboard
- MMSE / MoCA implementation
- 10+ NER languages
- 20 games
- Sophisticated telemedicine
- Massive cloud architecture
- Microservices
- Kubernetes

22. Planned Extensions

These are explicitly post-MVP. They are listed to show the intended direction, not as committed v1 work.

Extension 1 — Reinforcement-learning personalization

Once enough Track B data exists: current rule engine → contextual bandit → more sophisticated RL. It could choose which game, what difficulty, how long, and what content for each individual.

Extension 2 — Personalized content selection

Personalization doesn't have to control only difficulty. Eventually it could learn that (e.g.) User A engages more with family photographs, User B performs better with music-based activities, User C prefers visual matching — then select the next activity from user profile + past engagement + performance. That is more interesting than simply Level 4 → Level 5.

Extension 3 — NER voice assistant

Add Assamese speech → ASR → assistant → TTS → Assamese spoken response, then expand to other languages. This could become one of the project's strongest regional differentiators.

Extension 4 — Speech-based cognitive research

Later, study pause duration, speech rate, word-finding difficulty, repetition, response latency, and vocabulary diversity, and whether these change longitudinally. This needs proper clinical datasets and validation — research-stage.

Extension 5 — Clinician portal

Later, the clinician could see longitudinal charts, exported reports, activity history, and clinically relevant assessments, and potentially add verified assessment results.

Extension 6 — Clinical validation

App telemetry + clinician assessments + months of longitudinal data → a research dataset. Then study: can passive cognitive-game performance provide useful supplementary markers of cognitive change? That is a stronger research question than "can we build an AI that detects dementia?"

Extension 7 — Geofencing / safe-return

Later add a home safe-radius (e.g. 500m); if the person exits the radius, alert the caregiver. Technically feasible but not essential to proving the core cognitive-platform idea.

Extension 8 — More NER cultural packs

The architecture can eventually support per-region packs (Assam, Nagaland, Manipur, Mizoram, Meghalaya, Tripura, Arunachal, Sikkim), each containing language strings, images, foods, music, festivals, landmarks, and games. Instead of rebuilding the app for every region, you plug in different localized content packs.

23. Most Important Research / Data Constraint

No matter how good our model becomes, we should never casually claim "our gameplay predicts dementia." The scientifically responsible progression is:

```
                V1   Cognitive engagement
                          ↓
           V2   Personalized performance monitoring
                          ↓
        V3   Detect statistically meaningful deviation
                          ↓
      Clinical study — compare against validated assessments
                          ↓
 Only if evidence supports it: clinical decision-support research
```

Overclaiming disease-modifying or clinical effects is explicitly out of bounds for this project.

24. The Whole Project in One Flow

One elderly user opens the app:

> Good morning. Today is Wednesday.
>
> Would you like to play a memory game?

They play. Result: Accuracy 78%, Reaction time 3.4s, Difficulty 3. The result is stored locally. When internet becomes available:

```
SQLite  →  FastAPI  →  PostgreSQL
```

Their caregiver later opens the dashboard: 5 sessions this week; memory performance stable; reaction time stable. The next game gets adjusted based on recent performance. Over months, their data forms a personal history:

```
  Week 1 — Week 5 — Month 3 — Month 6
     |        |         |         |
accuracy   RT      errors   engagement
     └────────┴─────────┴─────────┘
                          ↓
             Personal baseline
                          ↓
         personalization / trends
```

Meanwhile the app gives them reminders, familiar photographs, regional memories, cognitive activities, and eventually spoken local-language interaction.

The architecture is intentionally designed so that the first version is genuinely useful without pretending we already possess a clinically validated AI model, while the data generated by the product gives us a path toward much stronger ML and research extensions later.

25. Background & Justification

This section records the evidence base behind the project — why the problem is real, what the clinical literature does and does not support, what already exists, and the specific gap this platform addresses. Figures are drawn from published sources cited inline; where a source predates the current date, it should be re-verified before external use.

25.1 Is the problem relevant?

National picture. The LASI-DAD nationwide study (Lee et al., *Alzheimer's & Dementia*, 2023) estimates dementia prevalence for adults aged 60+ in India at 7.4%, meaning about 8.8 million Indians older than 60 live with dementia. The burden is projected to roughly double — from 8.8 million in 2016 to an estimated 16.9 million by 2036.

Prevalence is not uniform. The same study found considerable cross-state variation, from a low of 4.5% in Delhi to a high of 11.0% in Jammu & Kashmir. Prevalence is higher among women (nearly double that of men), among rural populations, and among those with low or no formal education — all gradients that apply heavily to rural NER.

> **Key caution on data.** The NER was under-represented in the primary national dataset. Of the entire North-East, only Assam was included in LASI-DAD's detailed cognitive assessment; Sikkim had no state-level estimate at all. States such as Nagaland, Manipur, Meghalaya, Mizoram, Tripura and Arunachal Pradesh have very little dedicated dementia epidemiology. This data gap is itself part of the justification for an NER-focused project.

Why NER is a distinct problem, not "India minus metros"

- **Fast-ageing, out-migration-heavy demographics** — younger people migrate for work, leaving elderly relatives with thinner caregiving support.
- **Extreme linguistic diversity** — many languages and dialects (Assamese, Bodo, Khasi, Mizo, Manipuri/Meitei, Nagamese, Kokborok, and others). Almost every existing app is English- or Hindi-only.
- **Low awareness and high stigma** — nationally, low awareness and stigma are prevalent among both the public and health professionals; dementia is often dismissed as normal old age.
- **Sparse specialist care** — few neurologists/geriatricians and patchy connectivity in hill and border districts.
- **Culturally distinct memory anchors** — reminiscence relies on personally and culturally familiar images, music, festivals and food; generic Western content does not transfer.

One local asset worth noting: ARDSI (Alzheimer's & Related Disorders Society of India) has a Guwahati chapter that has run awareness activity in the region — a potential local partner for content validation and deployment.

25.2 What the clinical evidence actually says

This matters for setting honest expectations. The evidence-based position:

- **Prevention / early stage — promising.** The long-running ACTIVE trial found that cognitive speed training can reduce the risk of dementia; evidence suggests cognitive games are most effective when used prior to the onset of dementia. Framing the platform as early-intervention and engagement is the most defensible position.
- **Mild dementia — feasible and modestly beneficial.** A 24-week home RCT using a computerized cognitive-training app in Alzheimer's patients reported good adherence (~80%) and improvement over time in visual/auditory memory, attention, and arithmetic tasks.

- **Do not over-claim.** For established MCI, available evidence could not determine whether cognitive training prevents clinical dementia or maintains function. Position the tool as improving quality of life, engagement, mood and daily function — not as a cure or guaranteed disease-modifier.
- **Reminiscence & Cognitive Stimulation Therapy have real support** and pair well with games; digital reminiscence systems show benefit for people with mild dementia. This is the most culturally localizable feature for NER.
- **Design lesson.** Existing apps are often narrowly focused on cognitive games and lack diverse content (arts, physical activity, social interaction), which reduces long-term engagement and adherence. A holistic, multi-domain design is preferred.

25.3 Existing platforms

Global / clinical

- **Constant Therapy** — among the most clinically validated; targets dementia, aphasia, stroke, TBI. English-centric.
- **Lumosity, CogniFit, BrainHQ** — consumer brain-training; BrainHQ derives from the ACTIVE speed-training lineage.
- **Research-grade tablet/TV programs** — multi-domain training used in nursing-home studies.

India-specific (closest reference points)

- **DemClinic** (Nightingales Medical Trust + ARDSI Bangalore) — telemedicine, expert-led cognitive assessment platform to widen access to screening, diagnosis and care.
- **DemLink App** (NMT) — educates families of people with dementia and provides access to care and support via a mobile app.
- **Moving Pictures India** (NIMHANS) — digital caregiver-education media, trialled in English/Hindi/Kannada.

25.4 The gap this platform fills

None of the existing platforms are, together: (a) NER-language-localized, (b) built around NER cultural reminiscence content, and (c) offline-first for low-connectivity districts. That intersection — cognitive games + memory assistance + caregiver tools + localized reminiscence + offline-first + NER language support + longitudinal behavioral data — is the differentiation, and it aligns with the platform definition in Section 1.

25.5 Honest constraints to carry into any pitch

- A validated “dementia-detection AI” cannot be ethically or scientifically claimed without clinical partnership and IRB-approved data collection — a multi-year research effort, not an MVP feature.
- What can credibly ship fast: a well-designed, localized, offline-first cognitive-engagement and memory-assistance app with a simple adaptive engine — useful on its own, and the data foundation for the harder ML work later.
- The “AI screening” component is a research roadmap item, not a v1 deliverable, and should be presented as such.

25.6 Public datasets referenced for the research track

These are relevant only to the future speech/decline-detection work (Tracks B and C); none are required for the MVP.

- **DementiaBank (Pitt Corpus)** — the standard corpus: recordings + transcripts of the “Cookie Theft” picture-description task, with MMSE scores as labels. Free but requires a TalkBank access request.

- **ADReSS / ADReSSo** — cleaned, age/gender-balanced subsets of DementiaBank built to standardize model comparison for cognitive-decline detection.
- **MultiConAD / Taukadial** — multilingual conversational corpora useful for cross-lingual transfer approaches.
- **AI4Bharat / Bhashini** — open Indic-language ASR/TTS/translation models and APIs with expanding NER-language coverage; the practical starting point for any NER speech work (verify current coverage, it changes often).

> **Critical limitation.** None of these public corpora are in NER languages. They are useful for prototyping the pipeline, transfer learning, and validating feature extraction — not for a deployable NER-language model. Closing that gap requires a small, consented, ethically-approved regional corpus, as described in Sections 13–14. This limitation should be stated plainly rather than glossed over.

End of ground-truth document. This reflects only what has been finalized; any item marked "later," "future," or "extension" is explicitly out of MVP scope. Background figures in Section 25 are from published sources and should be re-verified before external use.